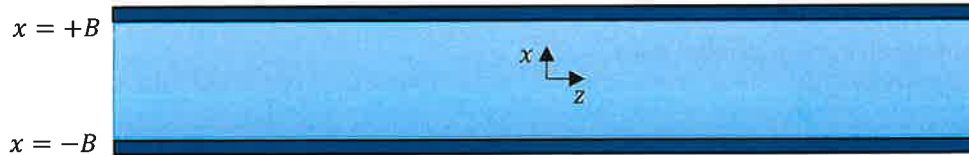


TRANSPORT PHENOMENA II, CHE 312, SPRING 2025

FINAL EXAM

CLOSED BOOK, TIME: 120 MINUTES



The above schematic and the reference frame apply to both problems.

1. Liquid B is laminarly flowing between two flat walls placed a distance $2B$ apart, with the velocity profile

$$v_z = v_{z,max} \left[1 - \left(\frac{x}{B} \right)^2 \right]$$

The solid walls are made of a species A that is slightly soluble in the liquid B . Species A diffuses into the liquid according to Fick's law for binary diffusion with a constant diffusivity D_{AB} , where it undergoes a chemical reaction according to a first order rate law $R_A = -k_1''' c_A$.

Answer the questions in terms of the following parameters (all assumed constant):

$v_{z,max}$: maximum velocity of the fluid in z -direction

B : half the distance between the two parallel walls

D_{AB} : Fick's law diffusivity of A in B

k_1''' : rate constant in units of inverse time.

- 1.1. By setting up a shell mole balance, derive a partial differential equation for the concentration of species A as a function of x and z . Assume the concentration of A is very small $x_A \ll 1$, and neglect the diffusive transport of A in z direction.

$$\int_V dV \frac{\partial}{\partial t} c_A = - \int_S dS \langle N_A | n \rangle + \int_V dV R_A$$

- 1.2. Sufficiently far downstream, the concentration profile c_A is independent of z . Find this asymptotic concentration profile as a function of x , assuming the concentration of A at both walls is c_{A0} .

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2. Very thick slurries and pastes sometimes move in a channel almost as a solid plug with a uniform velocity profile. Consider a fluid between two flat walls placed a distance $2B$ apart, with a uniform velocity profile $v_z = v_0$. For $z > 0$, heat is added to the fluid with a constant flux q_0 at both walls. The liquid enters the heated region of the slit at $z = 0$ with a uniform temperature T_0 .

Answer the questions in terms of the following parameters (all assumed constant):

v_0 : velocity of the fluid in z -direction

B : half the distance between the two parallel walls

k : thermal conductivity of the fluid

ρ : density of the fluid

c_p : isobaric heat capacity per unit mass of the fluid

q_0 : heat flux at the walls (assumed positive)

10/15

- 2.1. Write down a steady state macroscopic energy balance on the entire fluid between 0 and z to find the bulk average temperature as a function of z . Assume constant heat capacity and neglect the pressure dependence of enthalpy.

$$\int_V dV \frac{\partial}{\partial t} \rho \left(\frac{1}{2} v^2 + \phi + u \right) = - \int_S dS \rho \left(\frac{1}{2} v^2 + \phi + u \right) \langle v|n \rangle - \int_S dS \langle q|n \rangle - \int_S dS \langle v|\tau|n \rangle - \int_S dS p \langle v|n \rangle$$

$$T_b(z) \equiv \frac{\int_A dS v_z T}{\int_A dS v_z}$$

5/15

- 2.2. Write down a steady state shell energy balance to obtain a partial differential equation (PDE) for the temperature of the liquid as a function of x and z .

For large z , the temperature profile assumes the following asymptotic solution:

$$T(x, z) = T_b(z) + \psi(x)$$

0/15

- 2.3. Derive an ordinary differential equation (ODE) for $\psi(x)$ by replacing the asymptotic solution in the PDE derived in the previous part. Solve the ODE and evaluate the integration constants.

0/15

- 2.4. Using the asymptotic temperature profile derived above find the Nusselt number.

$$Nu = \frac{4hB}{k}$$

$$h[T(B, z) - T_b(z)] = q_0$$

Name:

$$1.1) 0 = - \int \delta S \langle N_A | \lambda \rangle + \int \delta V R_A, \quad N_{AX} = -D_{AB} \frac{\partial C_A}{\partial x}, \quad N_{AZ} = \frac{1}{1-x_A} = 1$$

$$0 = A_{ren} \cdot N_{AX}|_0 - A_{ren} \cdot N_{AX}|_{-B} + A_{ren} \cdot N_{AZ}|_Z - A_{ren} \cdot N_{AZ}|_{Z+BZ} - A_{ren} \delta x \delta z K_1''' C_A$$

$$0 = \frac{-\partial N_{AX}}{\partial x} - \frac{\partial N_{AZ}}{\partial z} - K_1''' C_A$$

$$0 = D_{AB} \frac{\partial^2 C_A}{\partial x^2} - V_{2,ren} + \frac{\partial C_A}{\partial z} - K_1''' C_A$$

$$1.2) 0 = D_{AB} \frac{\partial^2 C_A}{\partial x^2} - K_1''' C_A \rightarrow \frac{\partial^2 C_A}{\partial x^2} = \frac{K_1''' C_A}{D_{AB}}$$

$$\frac{\partial}{\partial x} C_A = \frac{K_1''' C_A}{D_{AB}} x + C_1 \rightarrow C_A = \frac{2 K_1''' C_A}{D_{AB}} x^2 + C_1 x + C_2$$

$$C_A \left(1 - \frac{2 K_1''' x^2}{D_{AB}} \right) = C_1 x + C_2$$

$$C_A = C_1 x + C_2, \quad X=B, \quad C_A = C_{A0}$$

$$X=-B, \quad C_A = C_{A0}$$

$$C_{A0} = -B C_1 + C_2 \rightarrow C_2 = C_{A0} + B C_1$$

$$C_{A0} = B C_1 + C_2 \rightarrow C_1 = 0 \rightarrow C_2 = C_{A0}$$

$$C_A = \frac{C_{A0}}{1 - \frac{2 K_1''' x^2}{D_{AB}}}$$

Name:

$$Z_{(1)} 0 = - \int_1 ds (q_1 \lambda) - \int_1 ds P V u$$

$$-A \cdot q_0 \cdot z + V_0 P (u_0 - u_1) = 0$$

$$q_0 - V_0 P C_P (T_1 - T_0) = 0$$

$$z \frac{q_0}{V_0 P C_P} + T_0 = T_1$$

$$\overline{T}_b = T_0 + \frac{\int_{-B}^B dx \int_0^z dz \frac{z q_0}{V_0 P C_P} \cdot V_{2, \max} [1 - (x/B)^2]}{\int_{-B}^B dx \int_0^z dz V_{2, \max} [1 - (x/B)^2]}$$

$$= T_0 + \frac{q_0}{V_0 P C_P} \cancel{V_{2, \max}} \int_{-B}^B dx \int_0^z dz z [1 - (x/B)^2]$$

$$\cancel{V_{2, \max}} \int_{-B}^B dx \int_0^z dz [1 - (x/B)^2]$$

$$\int_{-B}^B dx \int_0^z dz z - z(x/B)^2 \rightarrow \int_{-B}^B dx \frac{z^2}{2} - \frac{z^2}{2} (x/B)^2$$

$$\int_{-B}^B z - \frac{x^2}{B^2} dx$$

$$\hookrightarrow z x - \frac{x^3}{3 B^2} \Big|_{-B}^B$$

$$\hookrightarrow \left(z B - \frac{B}{3} \right) - \left(-z B + \frac{B}{3} \right)$$

$$\hookrightarrow 2 z B + \frac{2 B}{3}$$

$$\hookrightarrow \frac{z^2 x}{2} - \frac{z^2 x^2}{4 B^2} \Big|_{-B}^B$$

$$\hookrightarrow \left(\frac{z^2 B}{2} - \frac{z^2 B}{4} \right) - \left(-\frac{z^2 B}{2} + \frac{z^2 B}{4} \right)$$

$$\hookrightarrow z^2 B + \frac{z^2 B}{2}$$

Name:

$$\frac{q_0}{V_0 \rho C_P} \left(z^2 B + \frac{z^2 B}{2} \right) \rightarrow \frac{q_0 z^2 B + \frac{z^2 B}{2} q_0}{2 V_0 \rho C_P z B + \frac{2}{3} V_0 \rho C_P B}$$

$$2 z B + \frac{2 B}{3}$$

$$2.2) \frac{\partial u}{\partial z} - \frac{\partial q_0}{\partial x} = 0$$

$$\frac{du}{dz} + K \frac{\partial^2 T}{\partial x^2} = 0$$

$$2.3) \frac{du}{dz} + K \frac{\partial^2 T}{\partial x^2} = \frac{q_0 z^2 B + \frac{z^2 B}{2} q_0}{2 V_0 \rho C_P z B + \frac{2}{3} V_0 \rho C_P B} + \psi(x)$$

$0: \text{large } z$

$$\psi(x) = K \frac{\partial^2 T}{\partial x^2} - \frac{q_0 z^2 B + \frac{z^2 B}{2} q_0}{2 V_0 \rho C_P z B + \frac{2}{3} V_0 \rho C_P B}$$

Name:

$$2.4) h(4) = q_0 \rightarrow h = q_0 / 4$$

$$N_U = \frac{4hB}{K} \quad , \quad q_0 = -K \frac{\partial T}{\partial x}$$

$$N_U = \frac{4B}{K} \cdot \left(\frac{K}{B} \cdot \frac{4}{24} \right) = \boxed{\frac{16}{24}}$$

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